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$$f(x, y, z) = \sqrt{x^2 + y^2 + z^2} \text{ met } a(12, 3, 4)$$

$$\frac{\partial f}{\partial x} = \frac{x}{\sqrt{x^2 + y^2 + z^2}}$$

$$\frac{\partial f}{\partial y} = \frac{y}{\sqrt{x^2 + y^2 + z^2}}$$

$$\frac{\partial f}{\partial z} = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\frac{\partial f}{\partial x}(a) = \frac{12}{13}$$

$$\frac{\partial f}{\partial y}(a) = \frac{3}{13}$$

$$\frac{\partial f}{\partial z}(a) = \frac{4}{13}$$

$$\nabla f(a) = \left(\frac{12}{13}, \frac{3}{13}, \frac{4}{13} \right)$$

References